Hemodynamic responses during prolonged sitting.

Eight young men (group A) underwent 5 h of quiet sitting, preceded by 30 min of recumbency, 20 min of standing, and 20 s of walking, and five other young men (group B) underwent 70 min of sitting, preceded by recumbency only, to determine the effects of prolonged sitting and previous posture on hemodynamic responses (measured by impedance plethysmography). Group A showed more calf blood pooling and a decrease in thigh blood flow during sitting in comparison with the control group, but after 1 h of sitting hemodynamic responses of the two groups were similar. Sitting for 5 h (1st vs. 5th h) resulted in an increase in calf venous pooling (17%) and a decrease in calf BF (13%), a reduction in gravitational pooling in the thigh (corresponding to increased pooling in the calf), increases in diastolic and mean arterial pressures (6 and 7.3 mmHg, respectively), and minor changes in heart rate, stroke volume, and cardiac output. The results show that it is necessary to sit for 1 h before hemodynamic responses can be assessed in this position, regardless of the posture maintained previously. The main effect of prolonged sitting is pooling in the calf, which is compensated for by an increase in peripheral resistance.